

1

2

3

4

5

6

7

8

9

10

11

A

B

C

D

E

F

G

H

NATIVE SOURCE. Power development; safety/PHY/PCB/bench qualification remain OPEN.

01_carrier_interface

10_input_protection

20_positive_rails

30_negative_reference

40_power_supervision

03_peripheral_boundaries

04_dac_0_3

05_dac_4_7

06_analog_outputs

41_power_status

File: 01_carrier_interface.kicad_sch

File: 10_input_protection.kicad_sch

File: 20_positive_rails.kicad_sch

File: 30_negative_reference.kicad_sch

File: 40_power_supervision.kicad_sch

File: 03_peripheral_boundaries.kicad_sch

File: 04_dac_0_3.kicad_sch

File: 05_dac_4_7.kicad_sch

File: 06_analog_outputs.kicad_sch

File: 41_power_status.kicad_sch

Sheet: /

File: servohl_io_revB.kicad_sch

Title: ServoHIL Rev.B – native editable single-SoC expansion

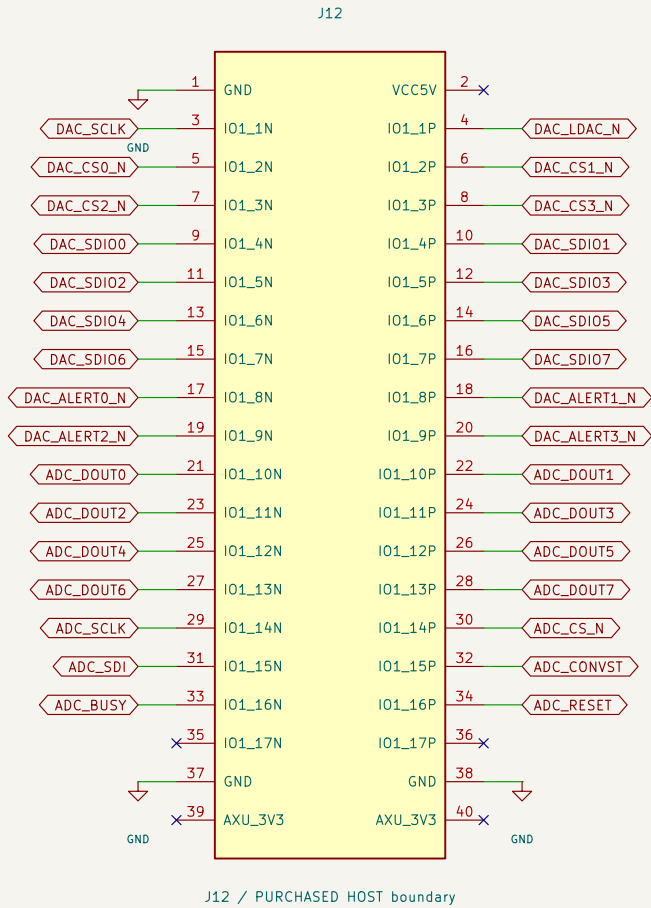
Size: A2

Date:

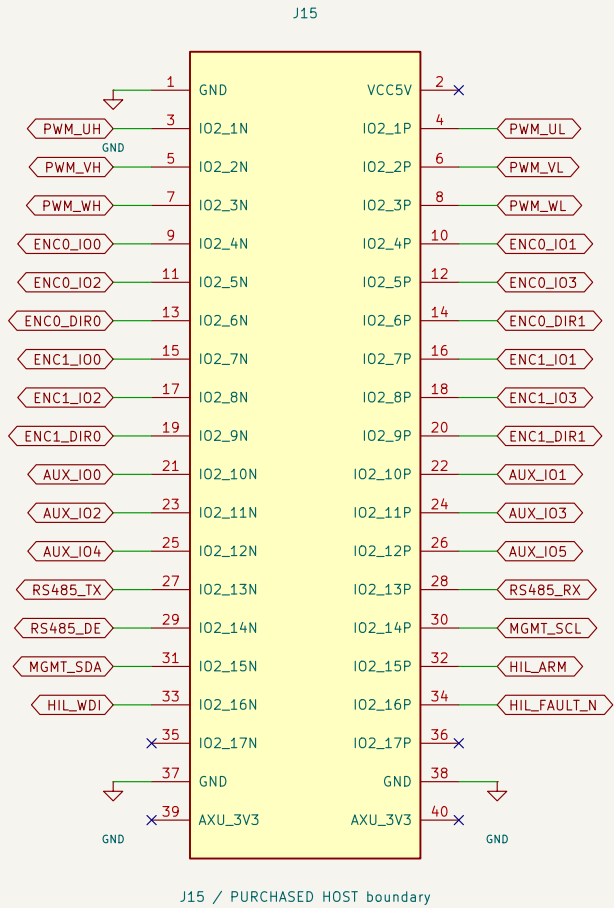
Rev: B-power-development

KiCad E.D.A. 10.0.6

Id: 1/11



J12 / PURCHASED HOST boundary



J15 / PURCHASED HOST boundary

Connector power contacts remain NC. 64 assigned GPIO; unused GPIO also NC. No secondary compute chip.

J12 and J15 are ORIGINAL AXU2CGB interfaces, not AXU2CGB-I or -E. No board-power backfeed through contacts.

Signal-pin backfeed, output enable and bank power sequencing STILL REQUIRE electrical review.

NON-RELEASE: physical power, safety, timing and package reviews remain open

magic-att/servoHIL

Sheet: /01_carrier_interface/

File: 01_carrier_interface.kicad_sch

Title: 01 carrier interface

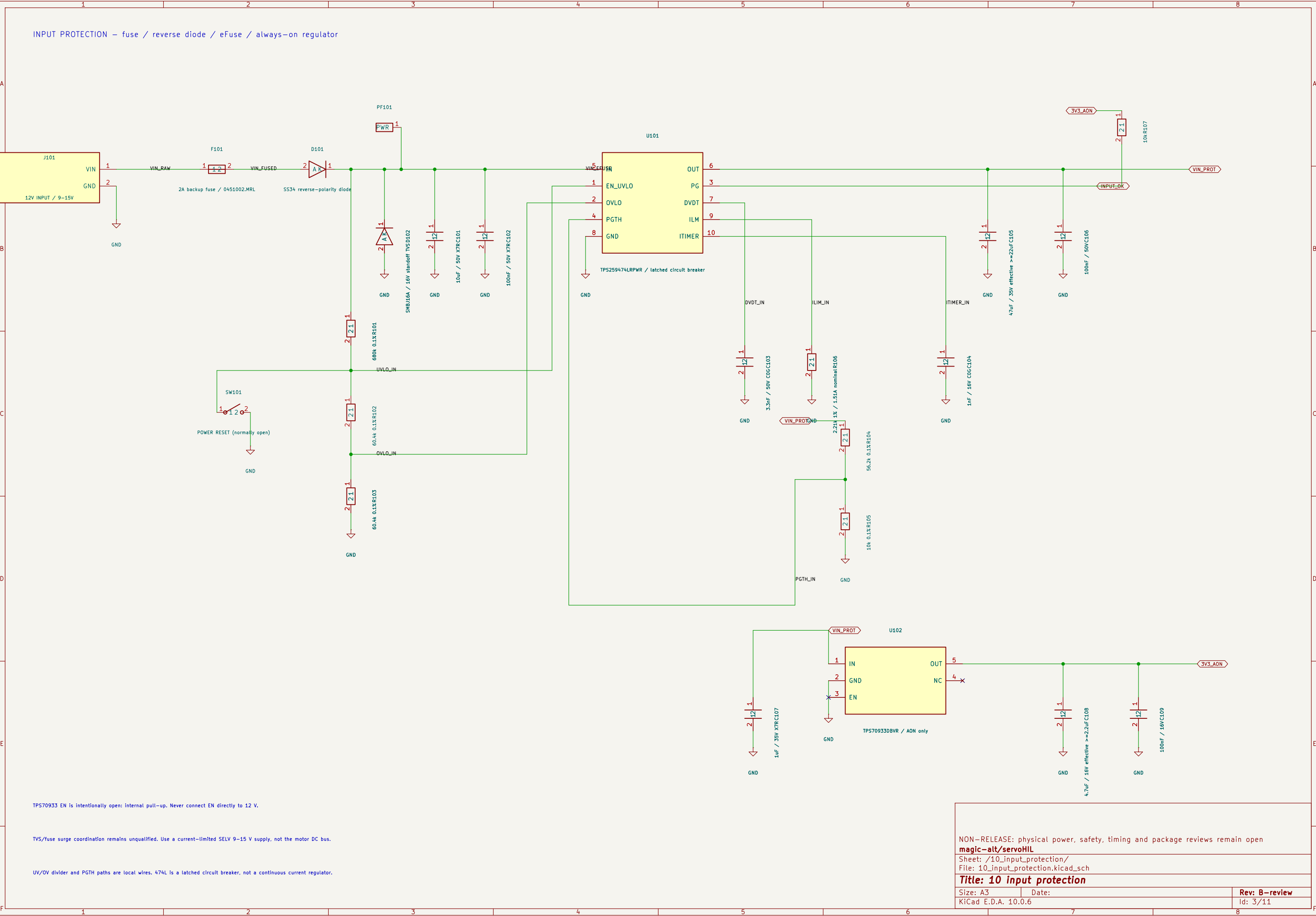
Size: A3

Date:

Rev: B-review

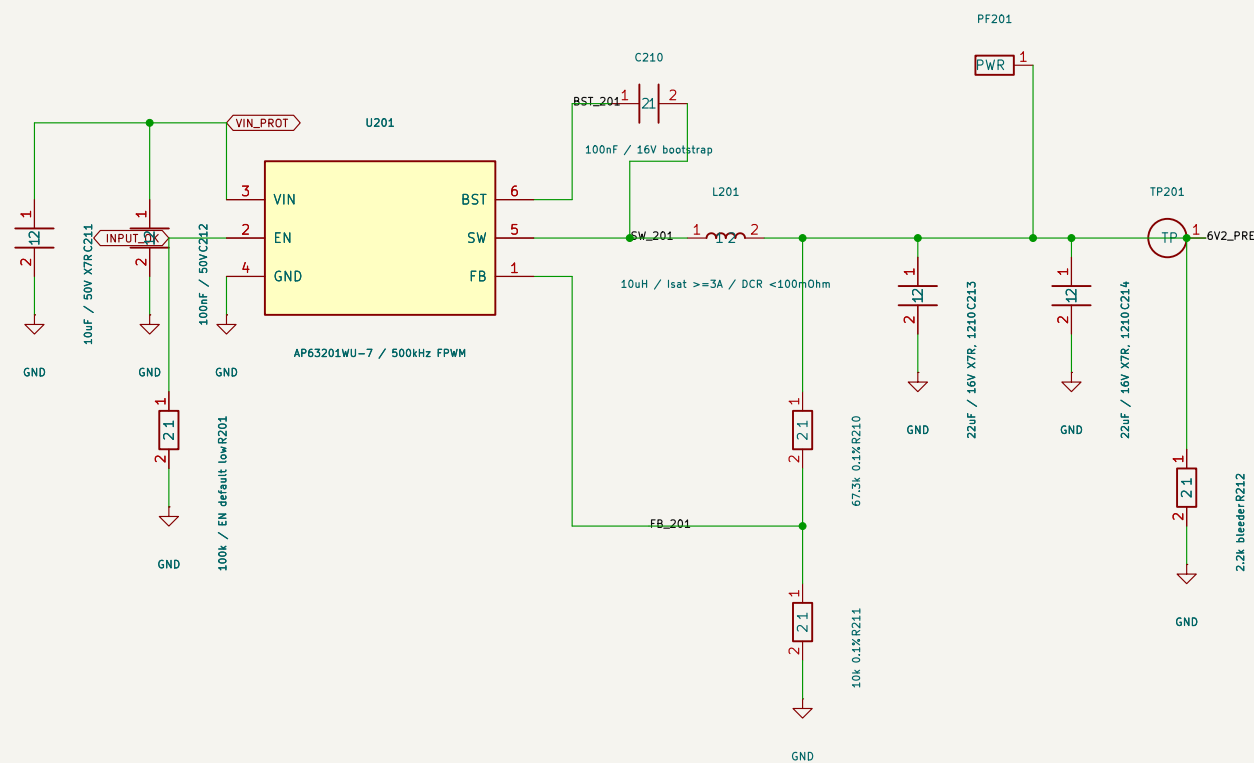
KiCad E.D.A. 10.0.6

Id: 2/11



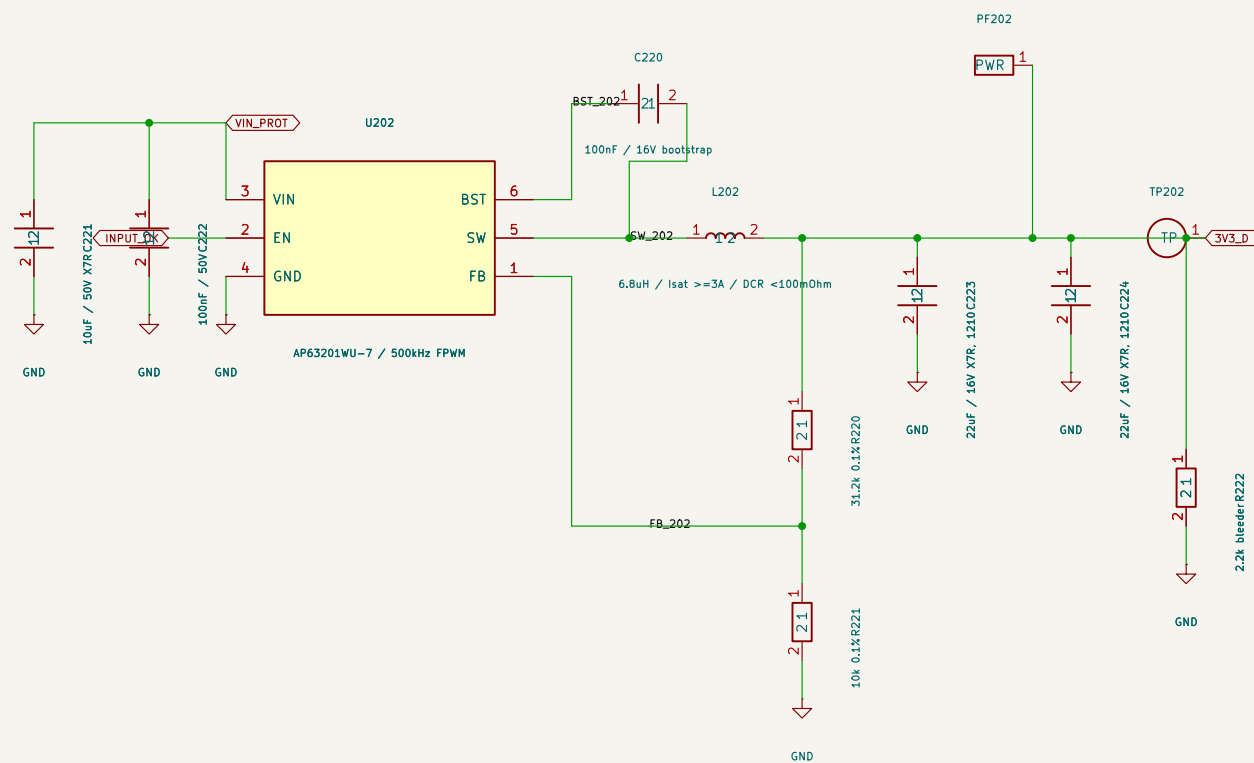
POSITIVE RAILS – direct power path, local feedback, cross-sheet interfaces only

6V2_PRE / AP63201, 500 kHz



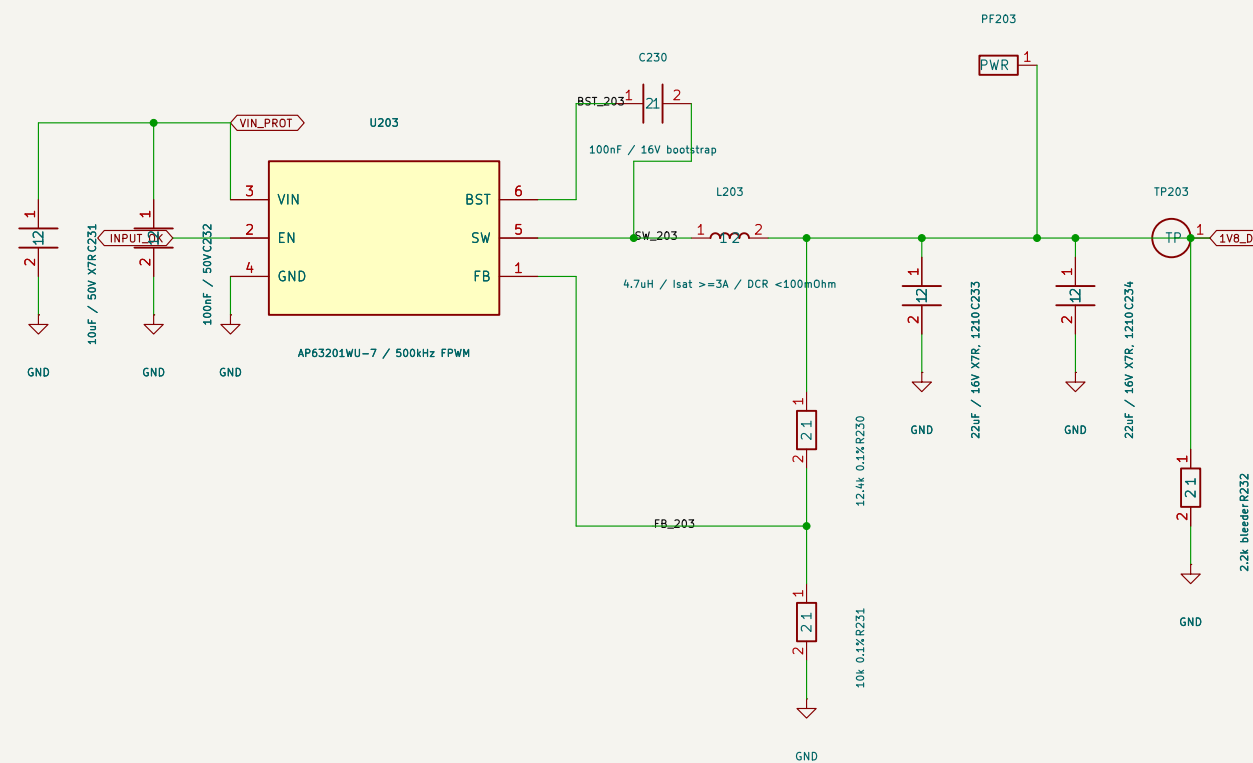
Cout effective capacitance $\geq 20 \mu\text{F}$; values/ratings unchanged.

3V3_D / AP63201, 500 kHz



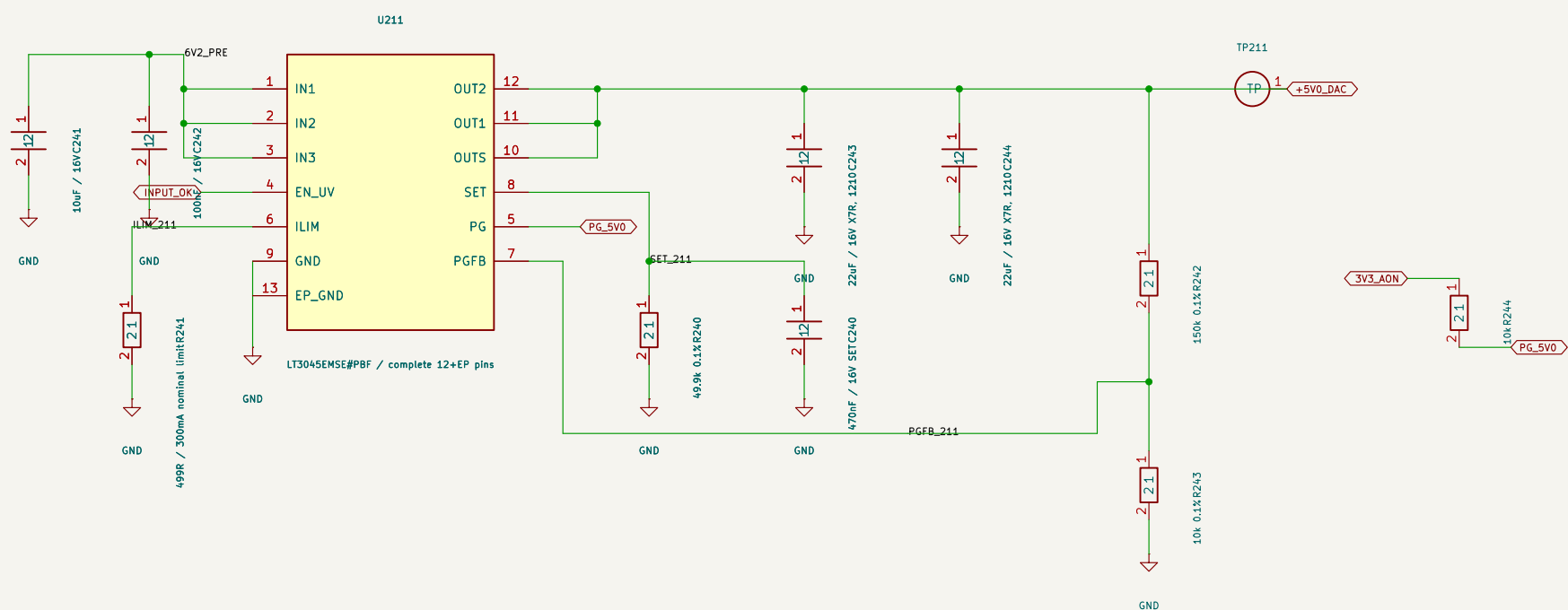
Cost effective capacitance $\geq 20 \mu\text{F}$; values/ratings unchanged.

1V8_D / AP63201, 500 kHz



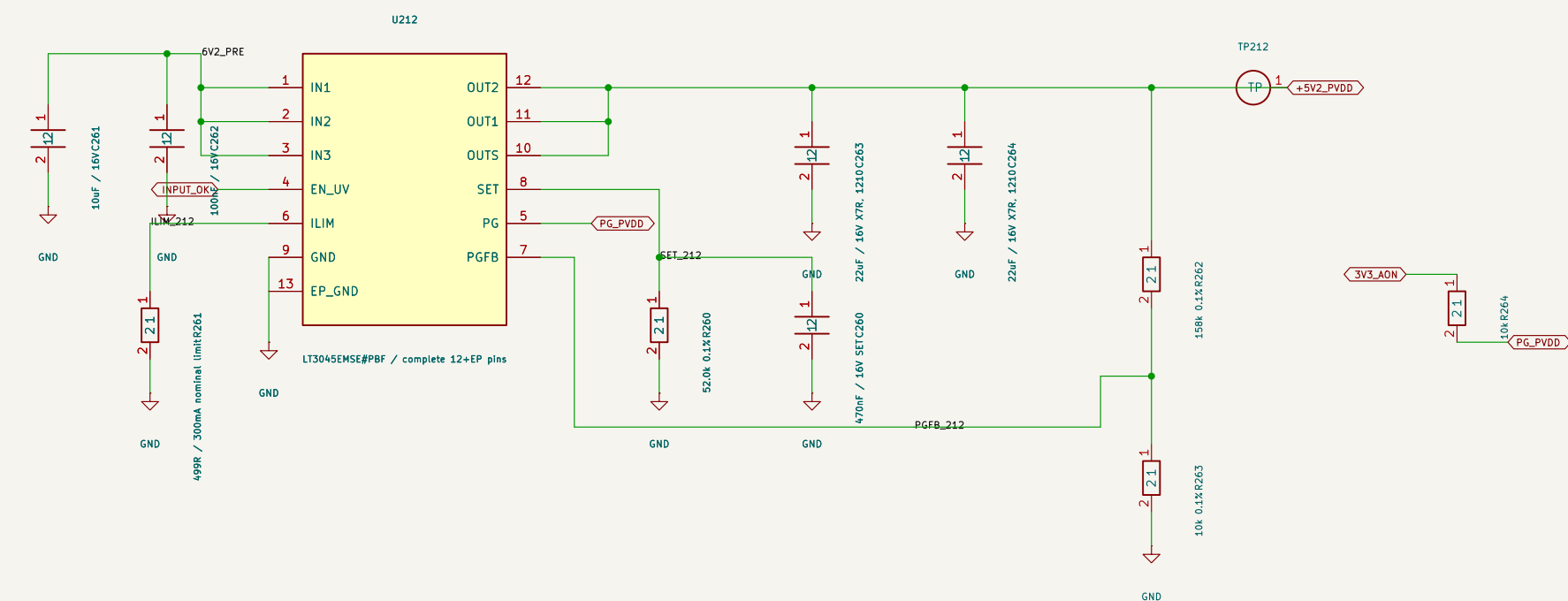
Cost effective capacitance $\geq 20 \mu\text{F}$; values/ratings unchanged

+5V0_DAC / LT3045 low-noise post-regulator



OUTS / SET Kelvin layout required. PG pull-up uses always-on 3.3 V.

+5V2_PVDD / LT3045 low-noise post-regulator



OUTS / SET Kelvin layout required. PG pull-up uses always-on 3.3 V.

All parts, pin numbers, values and footprints match the previous native circuit. These drawings do not establish bench qualification.

No second FPGA. Output current, capacitor DC bias, startup, ripple, thermal and safety behavior remain hardware review gates.

NON-RELEASE: physical power, safety, timing and package reviews remain open

magic-alt/servoHIL
Sheet: /20_positive_rails/
File: 20_positive_rails.kicad_sch

Title: 20 positive rails

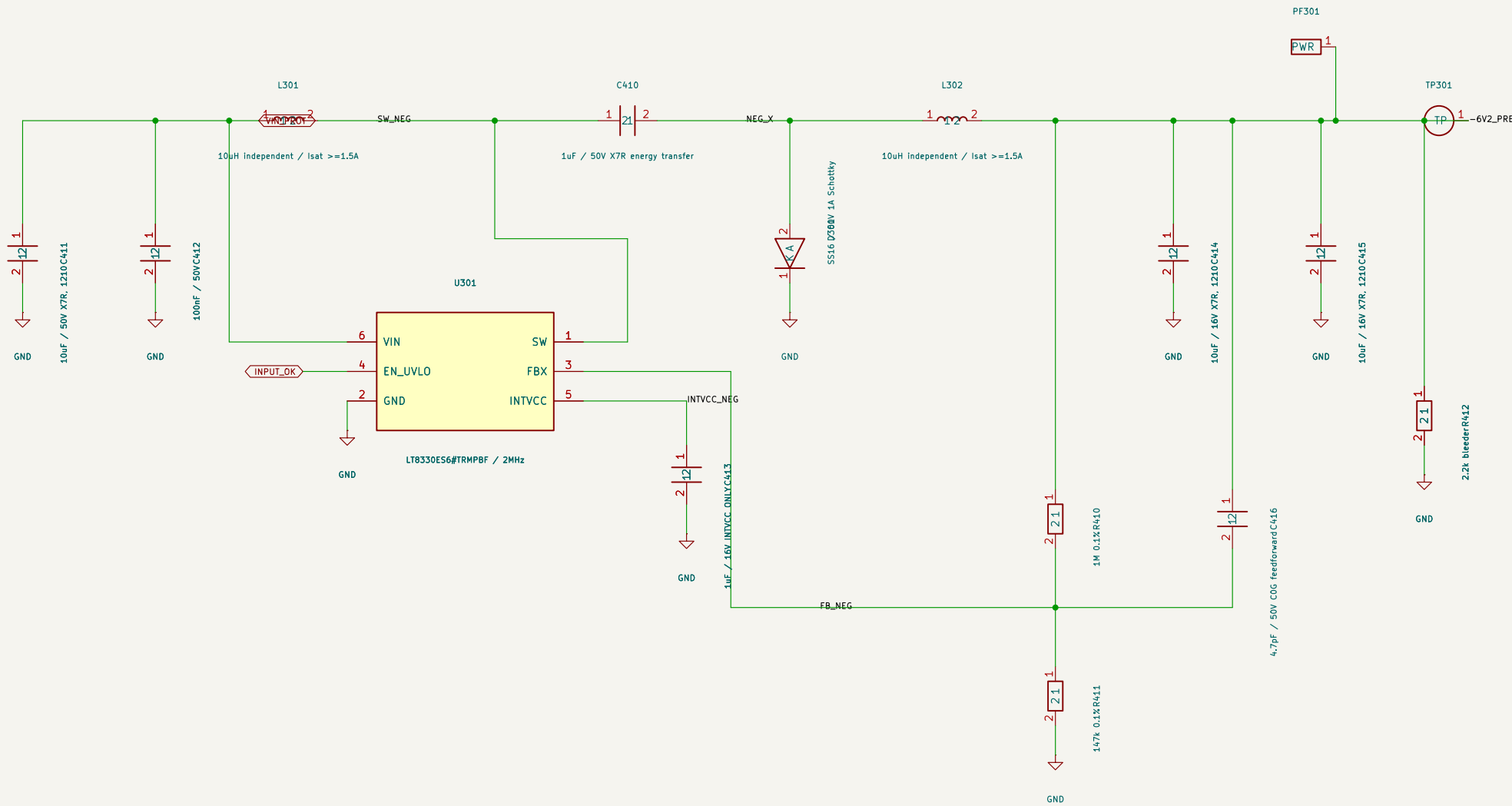
Size: A2	Date:
----------	-------

KiCad E.D.A. 10.0.6
10

Rev: B—review

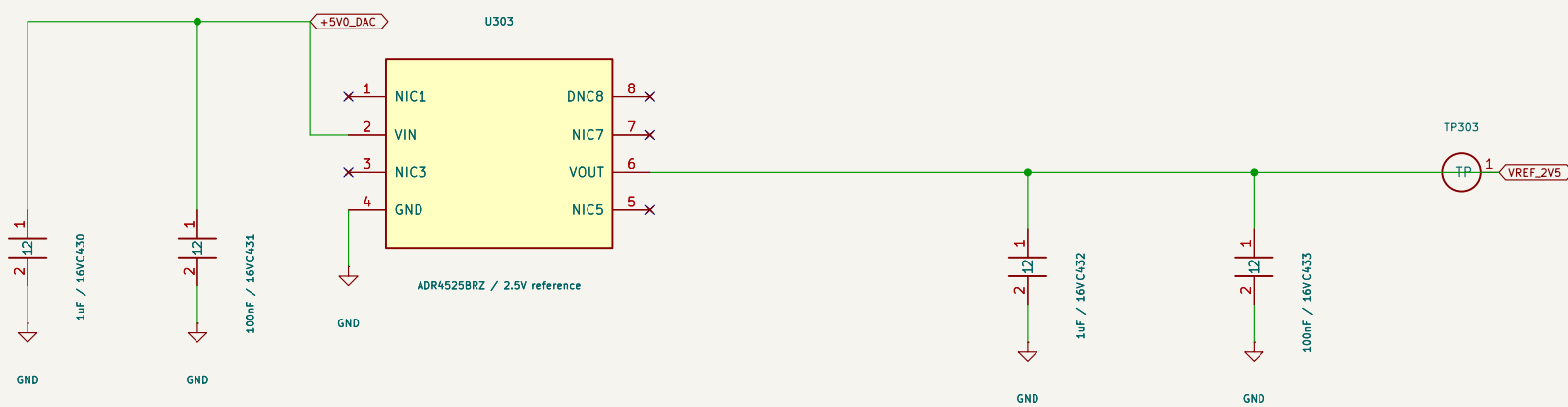
Id: 4/11

NEGATIVE RAIL / REFERENCE – wired Cuk converter, LT3094 and ADR4525



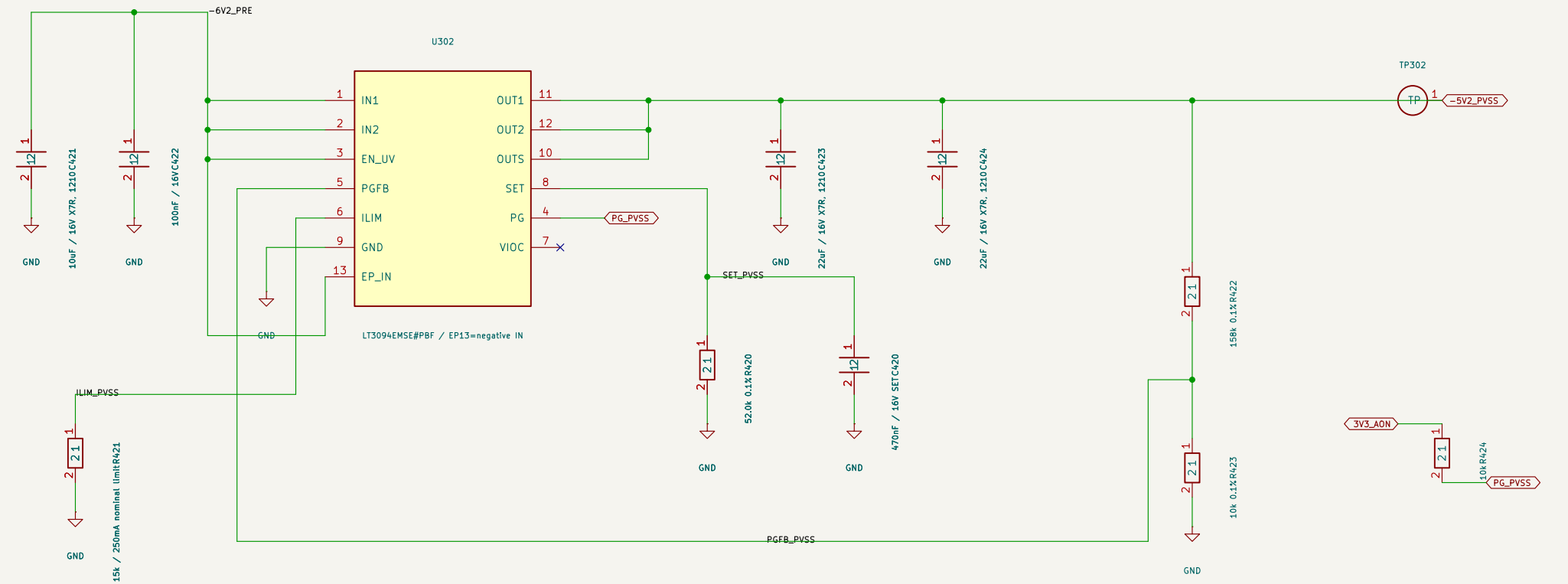
Cuk current path: VIN -> L301 -> C410 -> L302 -> negative output. D301 cathode is grounded.

Independent inductors; transfer capacitor sees $V_{IN} + |V_{OUT}|$. Existing component values are preserved.



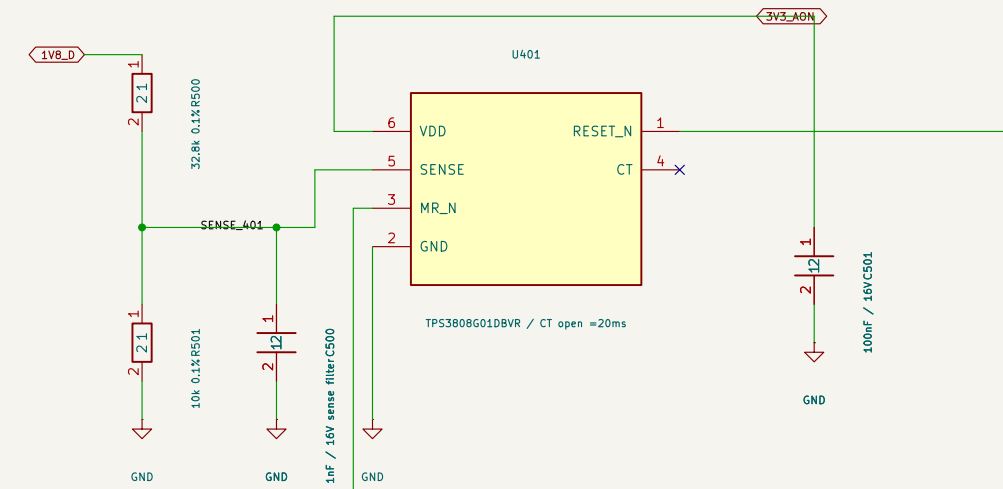
LT3094 EP13 / EN3 connect to negative IN. OUTS/SET require Kelvin layout. No startup/ripple/thermal PASS is implied.

ADR4525: unused NIC/DNC pads remain NC. DAC external—reference configuration and actual fanout/load require validation.

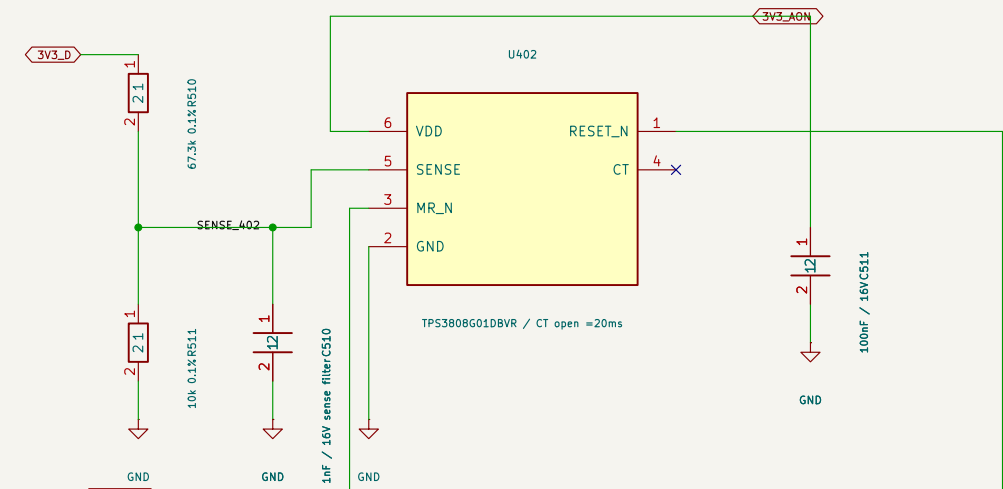


POWER QUALIFICATION – rail monitors, negative sense, deliberate re–arm and reset driver

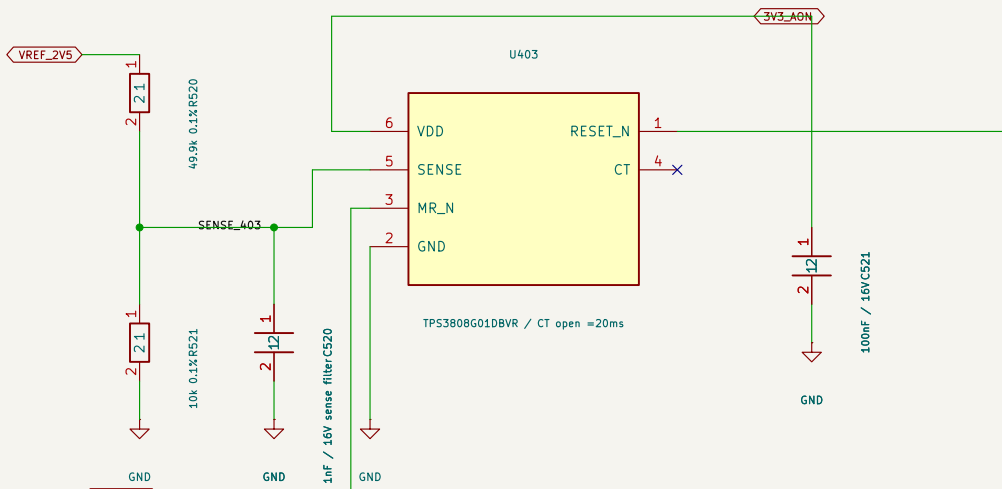
1V8_D rail supervision



3V3_D rail supervision

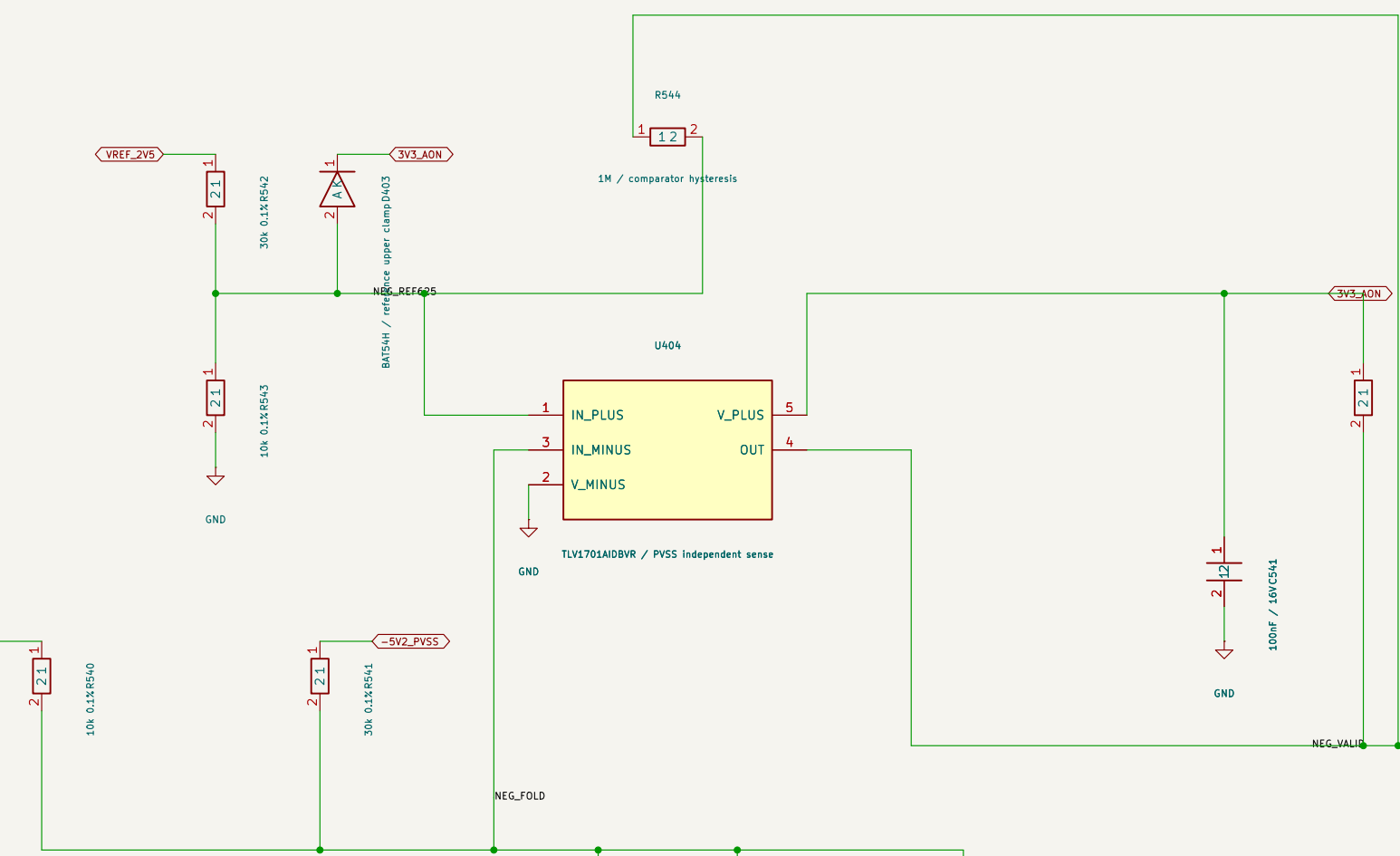


VREF_2V5 rail supervision

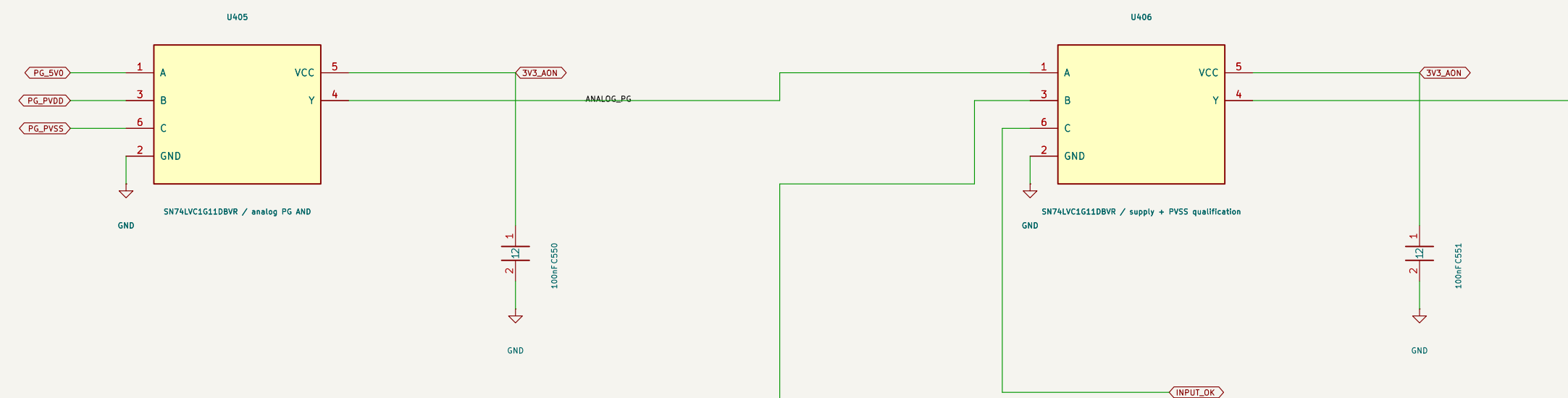


10k / open-drain pull-up R530

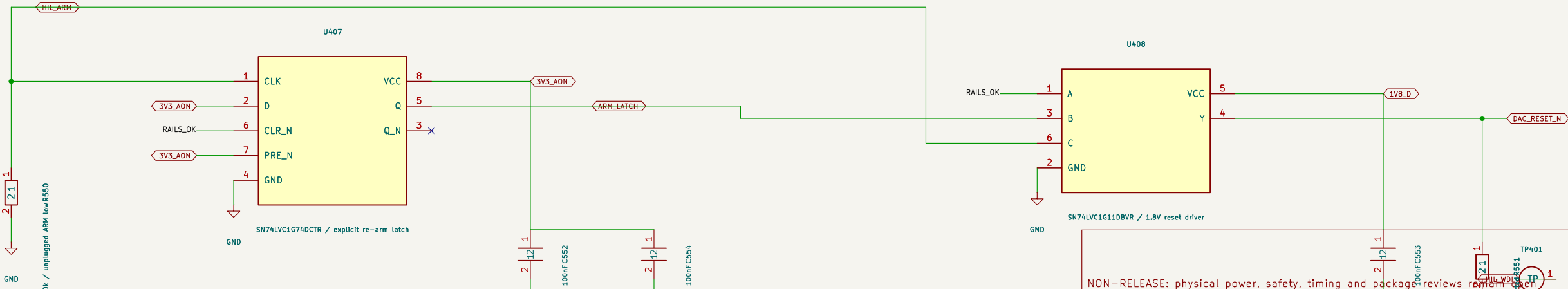
PVSS independent sense / hysteresis



Supply qualification



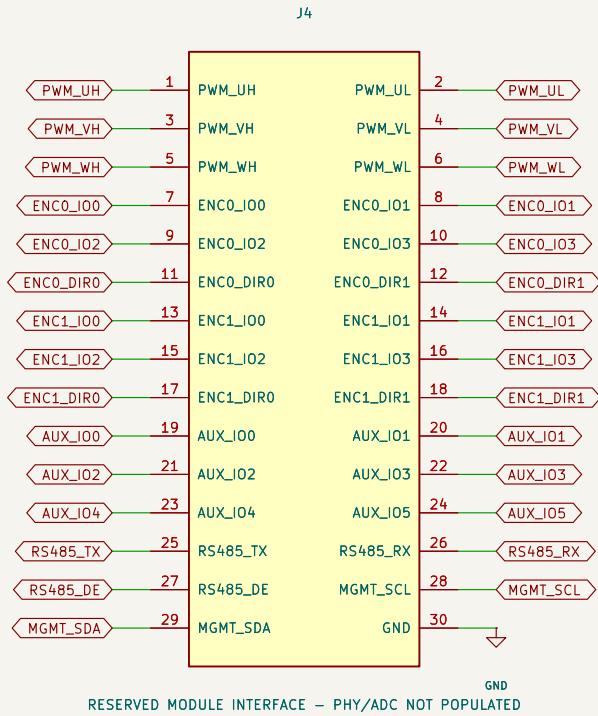
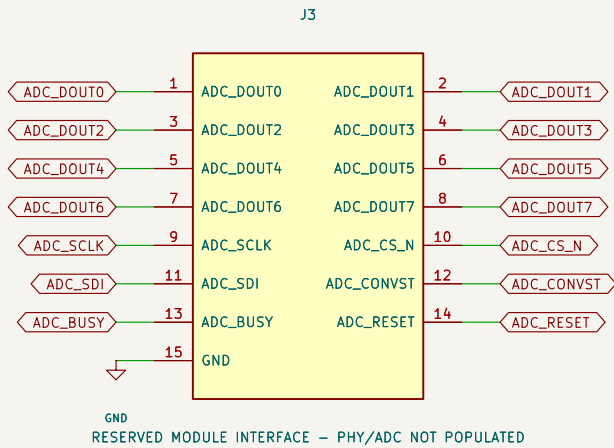
Re–arm latch / immediate disarm



TPA01_HIL_WDI is a reserved input, NOT a watchdog. Complete heartbeat, AO disconnect and DUT gate inhibit remain OPEN.

Power fault clears the latch: a fresh ARM edge is required. HIL_ARM LOW immediately holds DAC reset asserted. Hardware validation remains pending.

03 – ADC and digital PHY integration interfaces (reserved)



Eight AI channels are an interface reservation. No 1 MSPS verification is implied; 1.8V serial timing must be closed.

Each encoder uses one selected profile; ABZ/SSI/BISS/SPI are NOT simultaneously independent ports.

NON–RELEASE: physical power, safety, timing and package reviews remain open

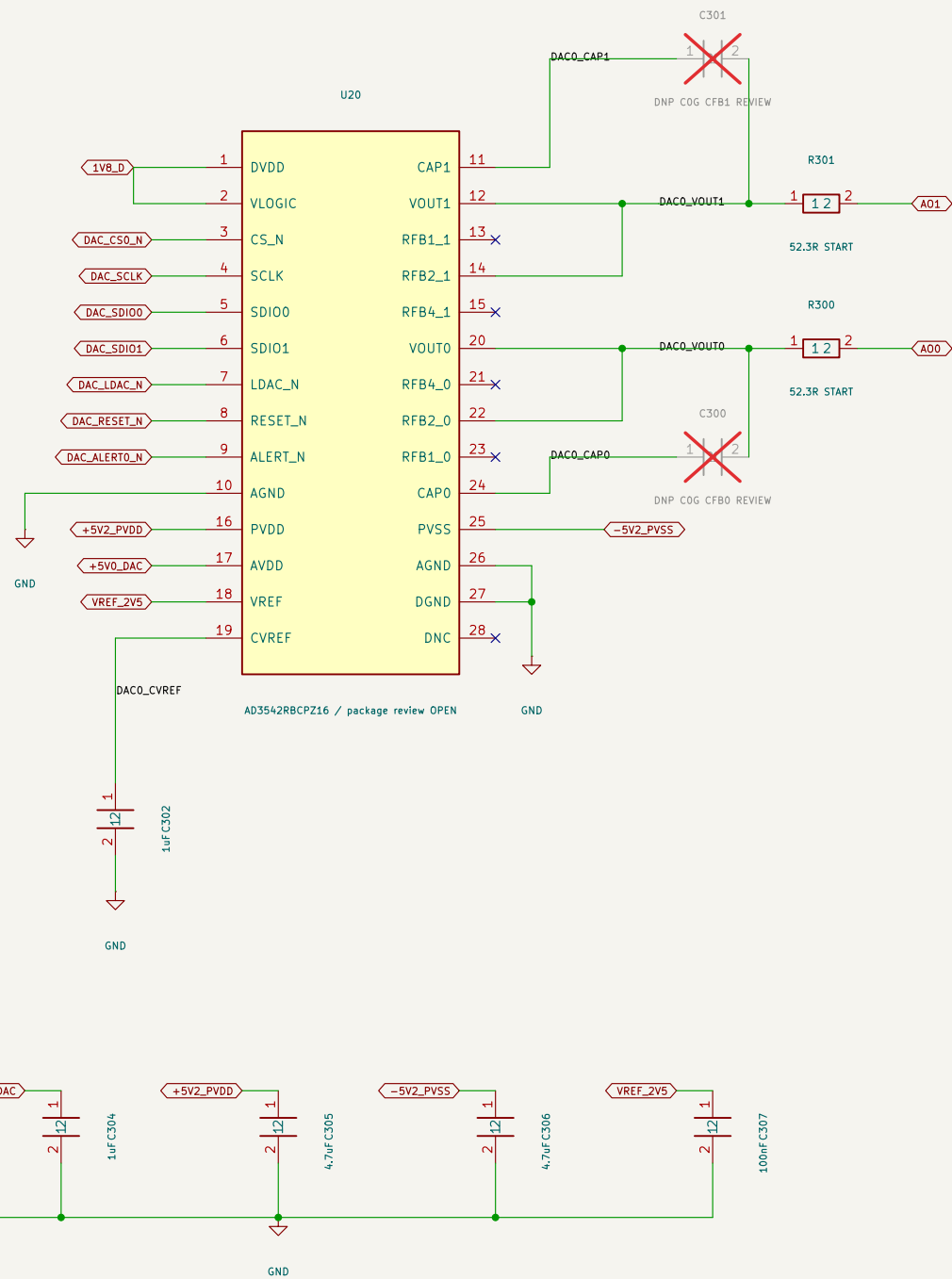
magic–alt/servoHIL

Sheet: /03_peripheral_boundaries/
File: 03_peripheral_boundaries.kicad_sch

Title: 03 peripheral boundaries

Size: A3	Date:	Rev: B–review
KiCad E.D.A. 10.0.6		Id: 7/11

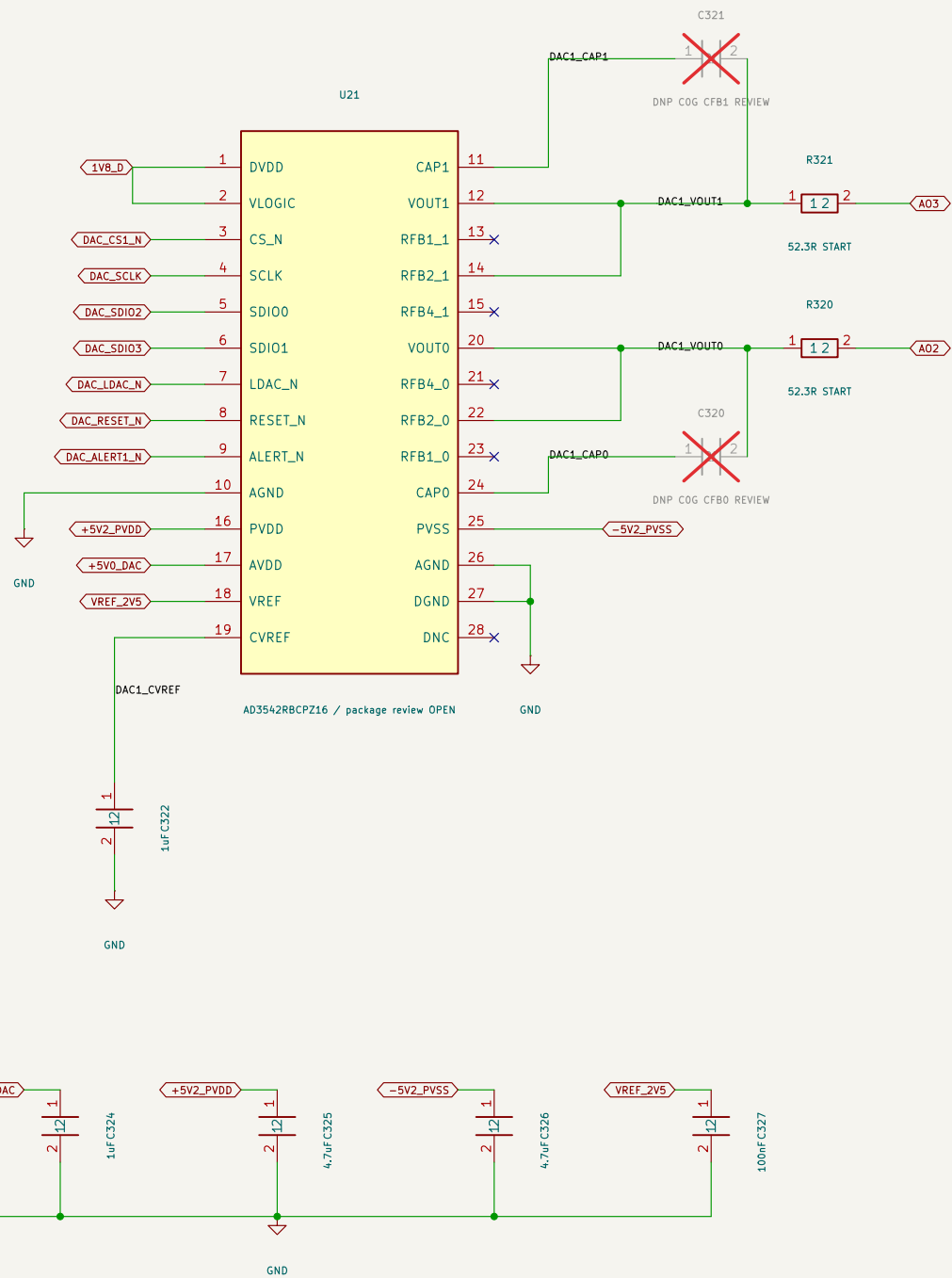
DAC – direct RFB2 feedback / CFB / output resistors; independent channels preserved



U20: A00 / A01. CFB remains DNP pending analog stability review.

Only inter-sheet signals use global labels. Feedback and output networks are drawn as visible closed local connections.

No electrical component/value change. CFB, output loading, package land pattern and hardware-safe AO behavior remain OPEN.



U21: A02 / A03. CFB remains DNP pending analog stability review.

NON-RELEASE: physical power, safety, timing and package reviews remain open

magic-att/servoHIL

Sheet: /04_dac_0_3/

File: 04_dac_0_3.kicad_sch

Title: 04 dac 0 3

Size: A3

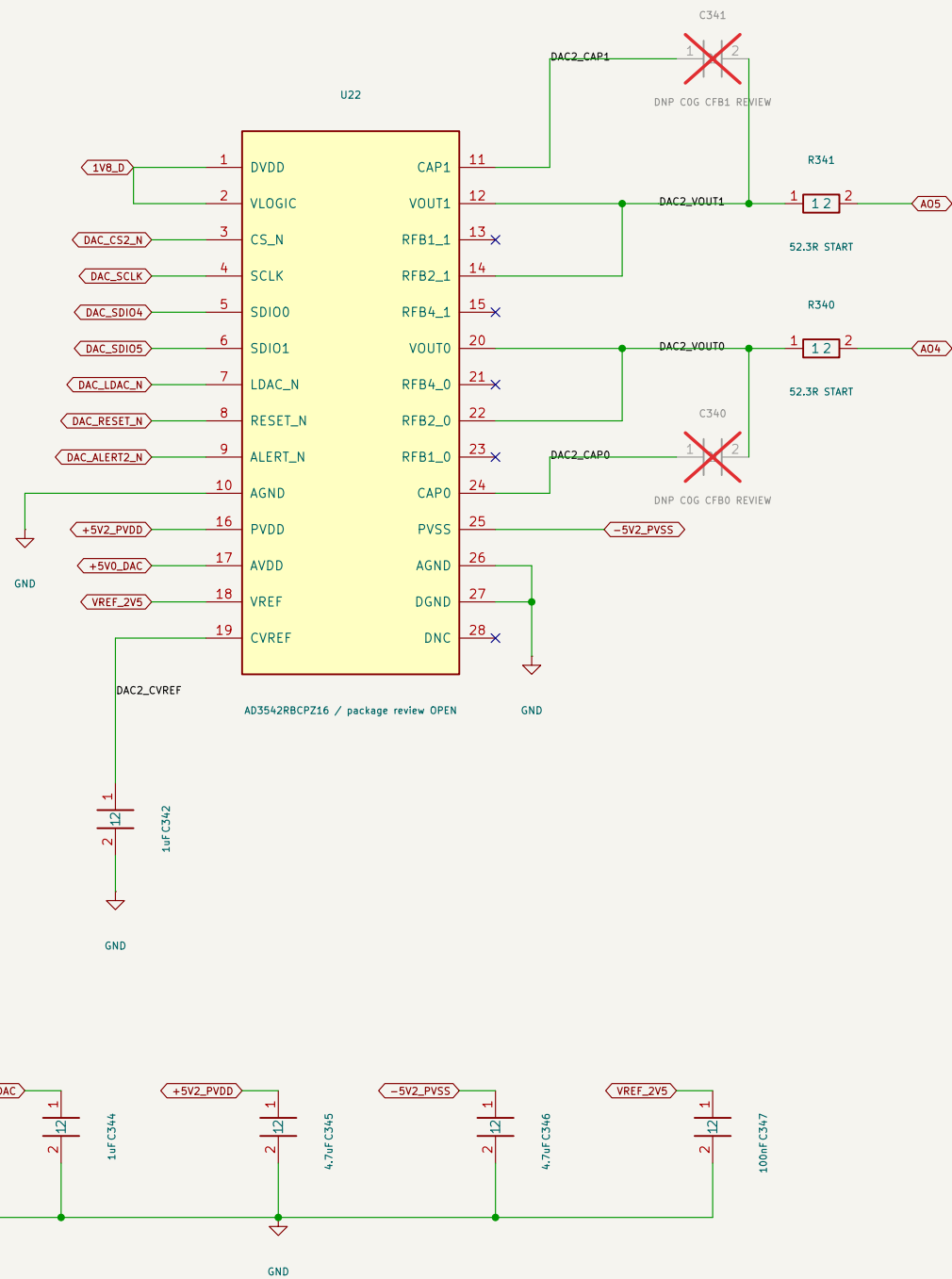
Date:

Rev: B-review

KiCad E.D.A. 10.0.6

Id: 8/11

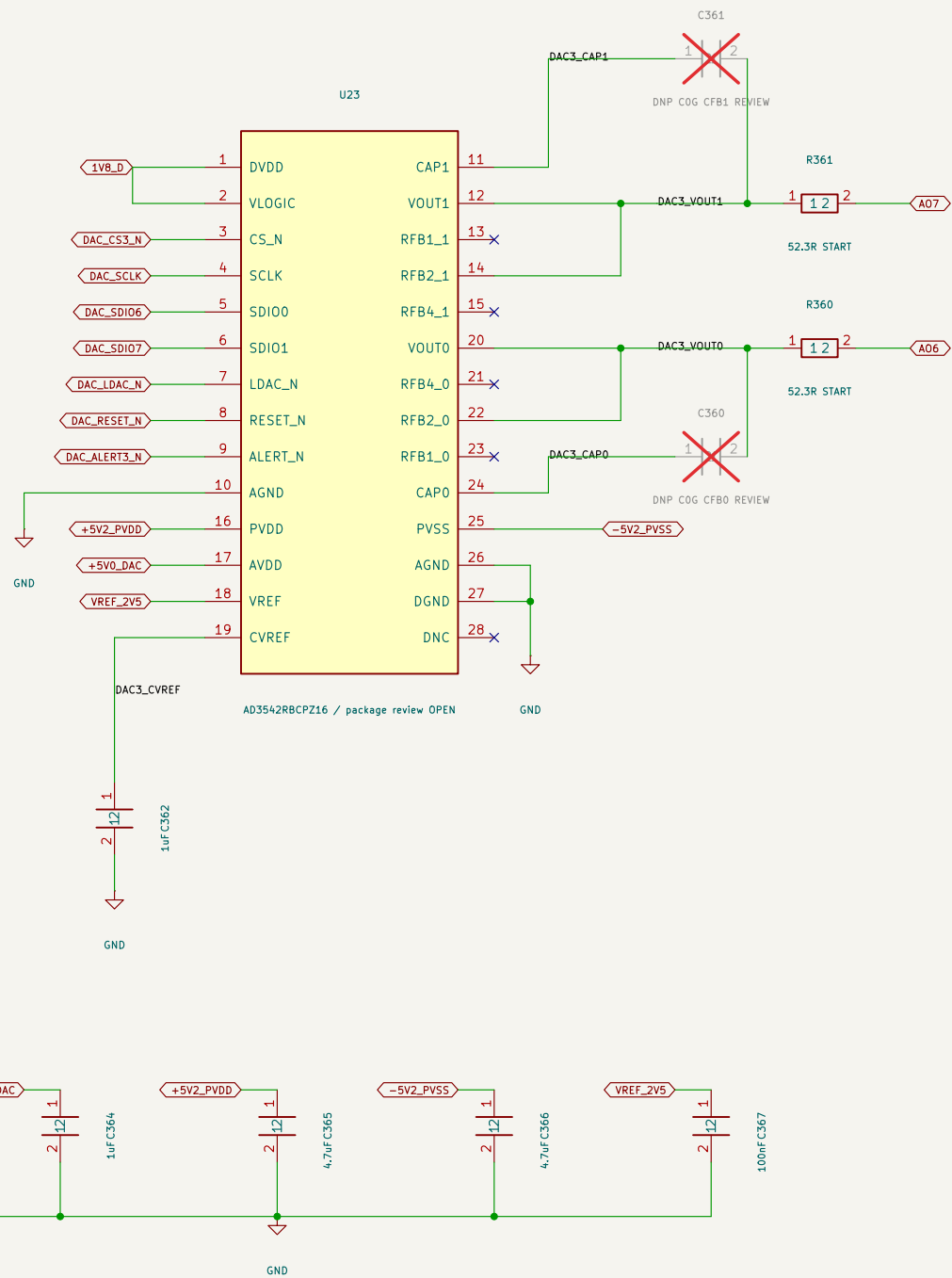
DAC – direct RFB2 feedback / CFB / output resistors; independent channels preserved



U22: A04 / A05. CFB remains DNP pending analog stability review.

Only inter-sheet signals use global labels. Feedback and output networks are drawn as visible closed local connections.

No electrical component/value change. CFB, output loading, package land pattern and hardware-safe AO behavior remain OPEN.



U23: A06 / A07. CFB remains DNP pending analog stability review.

NON-RELEASE: physical power, safety, timing and package reviews remain open

magic-att/servoHIL

Sheet: /05_dac_4_7/

File: 05_dac_4_7.kicad_sch

Title: 05 dac 4 7

Size: A3

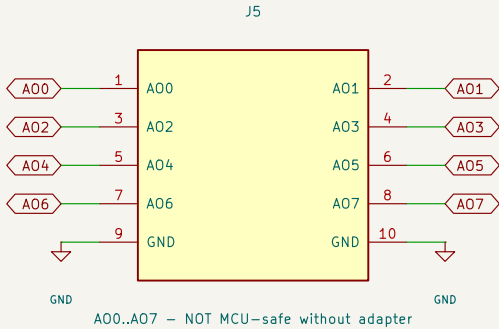
Date:

Rev: B-review

KiCad E.D.A. 10.0.6

Id: 9/11

06 – Eight analog outputs to reviewed DUT adapter



Default metadata: Ia, Ib, Ic, Vbus, torque, temperature, AUX0, AUX1. Electrical channels are equivalent.

DUT adapter supplies scaling/clamping and hardware inhibition. No direct +/-5V connection to MCU ADC.

